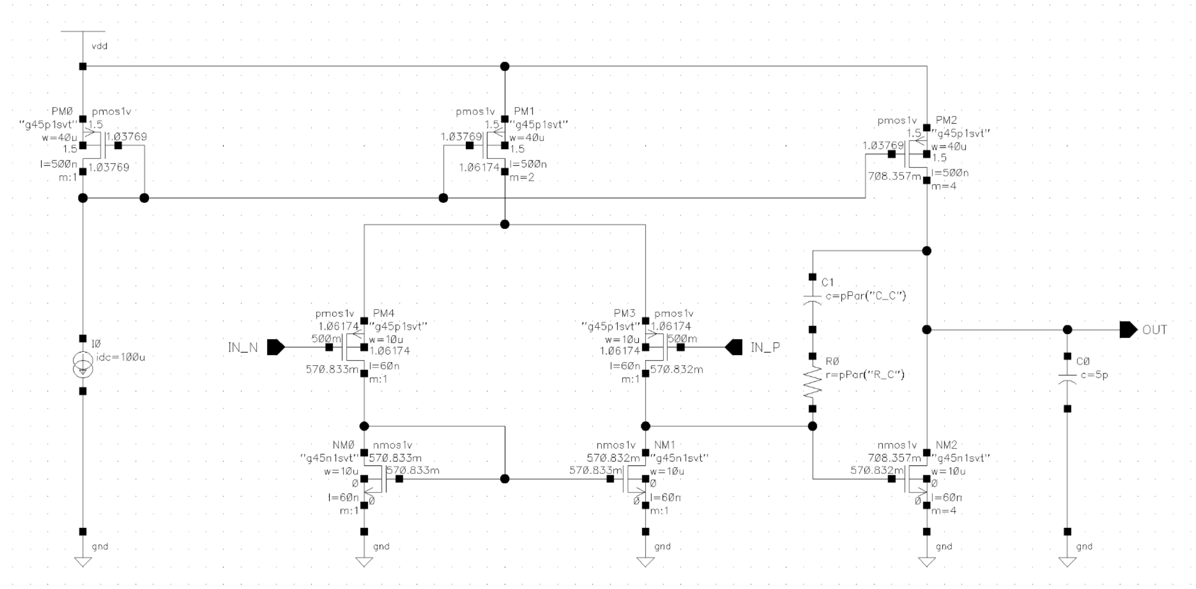
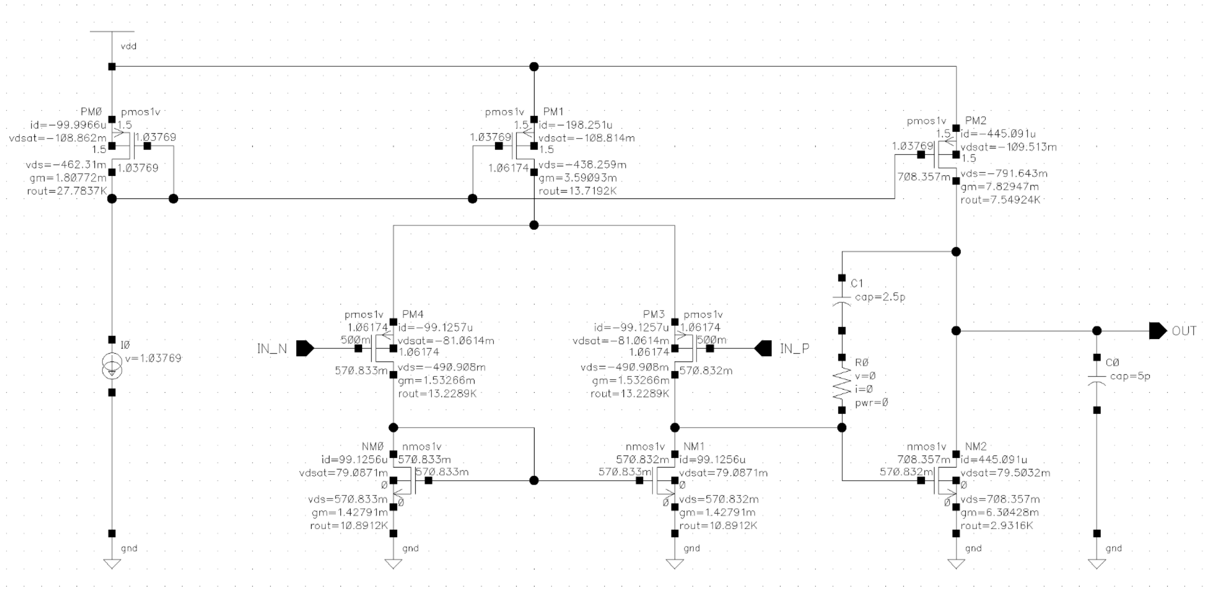
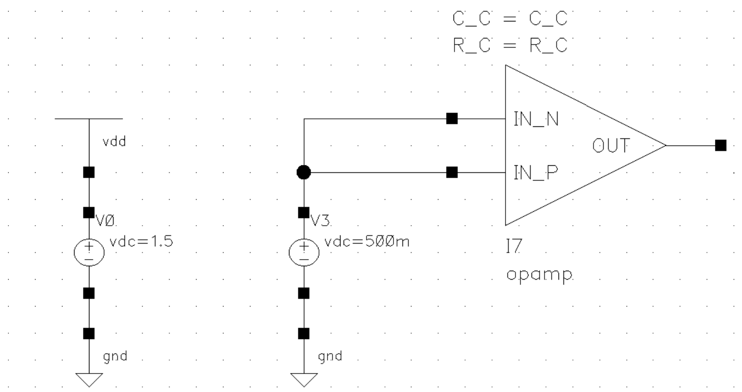
Chai Zi Yang A0262349Y EE5507 Tut 3

1. Estimate the small-signal differential voltage gain

The circuit is replicated as shown:







With the M1 M2 gate bias of 0.5 V the small signal parameters obtained from DC operating point simulation are summarised below:

g_{m2} (mS)	g_{m6} (mS)	r_{o2} (k□)	r_{o4} (k□)	r_{o6} (k□)	r_{o9} (k□)
1.53266	6.30428	13.2289	10.8912	2.9316	7.54924

*Note the MOSFET index follows the diagram given in the tutorial pdf, not Cadence

The small-signal differential voltage gain of the op-amp is then:

$$A_d = g_{m2} g_{m6} (r_{o2} || r_{o4})(r_{o6} || r_{o9})$$

$$A_d \approx 1.53266 \cdot 6.30428 \cdot (13.2289||10.8912) \cdot (2.9316||7.54924)$$

$$A_d \approx 1.53266 \cdot 6.30428 \cdot 5.9734 \cdot 2.1116$$

$$A_d \approx 121.9\, V/V$$

$$A_d \approx 41.7\, dB$$

2. Estimate the input CM range

Similarly, the relevant parameters obtained from DC operating point simulation are summarised below:

V_{DD} (V)	$\|V_{OD8}\|$ (V)	$\|V_{GS2}\|$ (V)	V_{D2} (V)	$\|V_{TH2}\|$ (V)
1.5	0.109	0.562	0.571	0.548

*Note the MOSFET index follows the diagram given in the tutorial pdf, not Cadence

Maximum input CM is limited by the M8 saturation:

$$V_{in,\,max} = V_{DD} - |V_{OD8}| - |V_{GS2}|$$

$$V_{in,\,max} \approx 1.5 - 0.109 - 0.562$$

$$V_{in,\,max} \approx 0.829\, V$$

Minimum input CM is limited by the M2 saturation:

$$|V_{DS2}| > |V_{GS2}| - |V_{TH2}|$$

$$V_{SD2} > V_{SG2} - |V_{TH2}|$$

$$-V_{D2} > \ -V_{G2} - |V_{TH2}|$$

$$V_{G2} > V_{D2} - |V_{TH2}|$$

$$V_{in,\,min} = V_{D2} - |V_{TH2}|$$

$$V_{in,\,min} \approx 0.571 - 0.548 \approx 0.023\, V$$

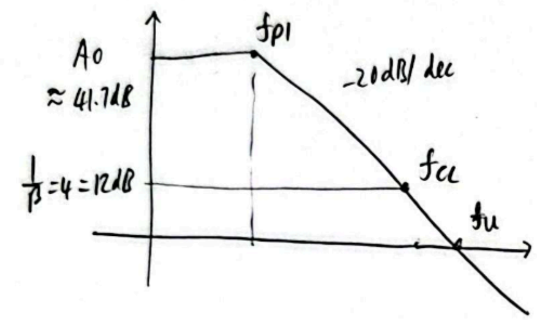
The range of input CM is then:

$$V_{CM,\,range} = V_{in,\,max} - V_{in,\,min} \approx 0.829 - 0.023 = 0.806\, V$$

3. Estimate C_C and R_C

If pole - zero cancellation is done

=> Assume system is single pole => max phase is $-90°$ => PM > 75° is guaranteed



$$20\log\left(\frac{f_{cL}}{f_{p1}}\right) = 41.7 - 12$$

$$\frac{f_{cL}}{f_{p1}} = 30.55$$

$$20\log\left(\frac{f_u}{f_{p1}}\right) = 41.7$$

$$\frac{f_u}{f_{p1}} = 121.62$$

$$\angle A(f_{cL}) = 0° - \tan^{-1}\left(\frac{f_{cL}}{f_{p1}}\right)$$

$$= -\tan^{-1}(30.55)$$

$$= -88.1° \quad \Rightarrow PM = 91.9° > 75° \ \#$$

Since there is no requirement on the PM for other values of β nor what the unity bandwidth is,

let $C_c = 2.5pF$ as an initial guess

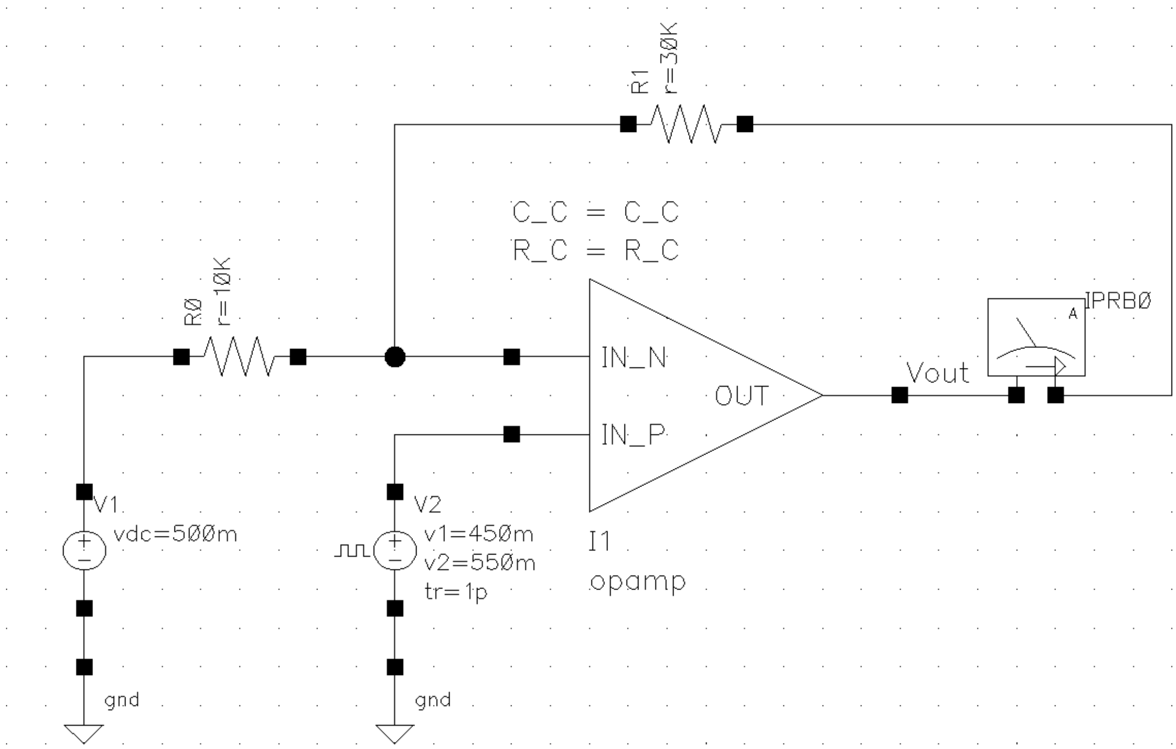
$$\Rightarrow f_u \approx \frac{1}{2\pi}\frac{g_{m2}}{C_c} \approx \frac{1}{2\pi} \times \frac{1.53266mS}{2.5pF} \approx 97.6\,MHz$$

for pole - zero cancellation

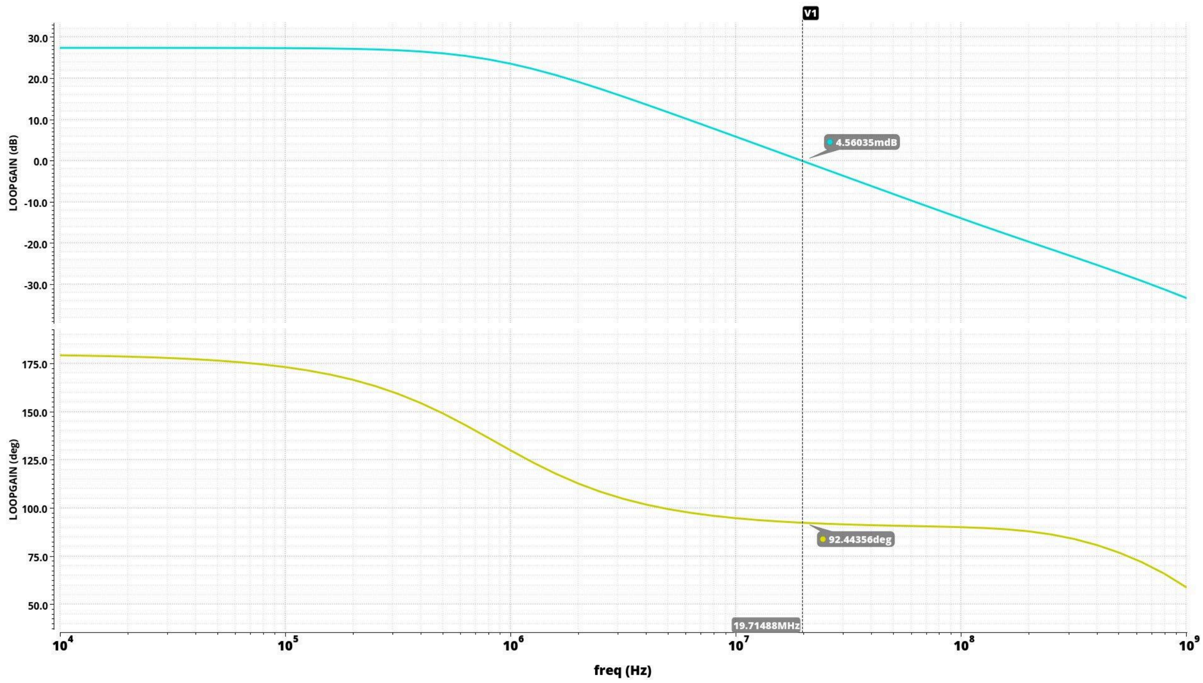
$$\Rightarrow R_c = \frac{1}{g_{m6}}\left(\frac{C_L + C_c}{C_c}\right) \approx \frac{1}{6.30428mS}\left(\frac{5+2.5}{2.5}\right) \approx 475.9\,\Omega$$

$\therefore$ $C_c = 2.5pF$ & $R_c = 475.9\Omega$

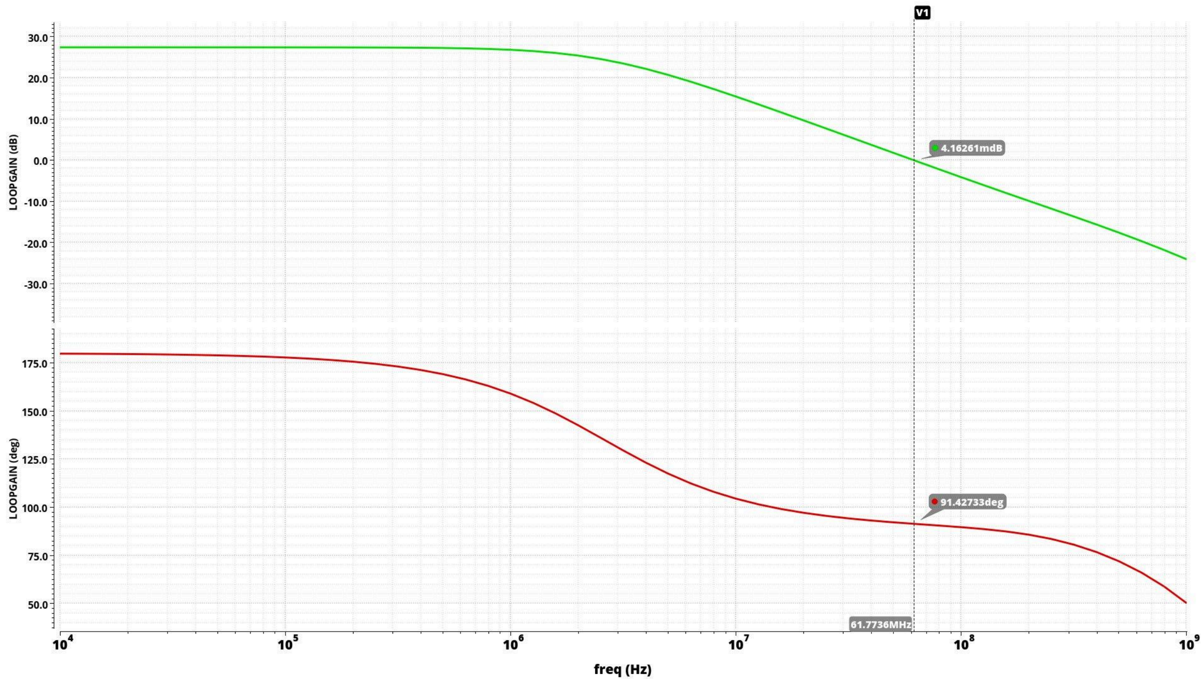
4. Verify your design using VIRTUOSO



Using the calculated C_C and R_C of 2.5 pF and 475.9 □, the loop gain and the closed loop AC frequency response are simulated with the circuit above.

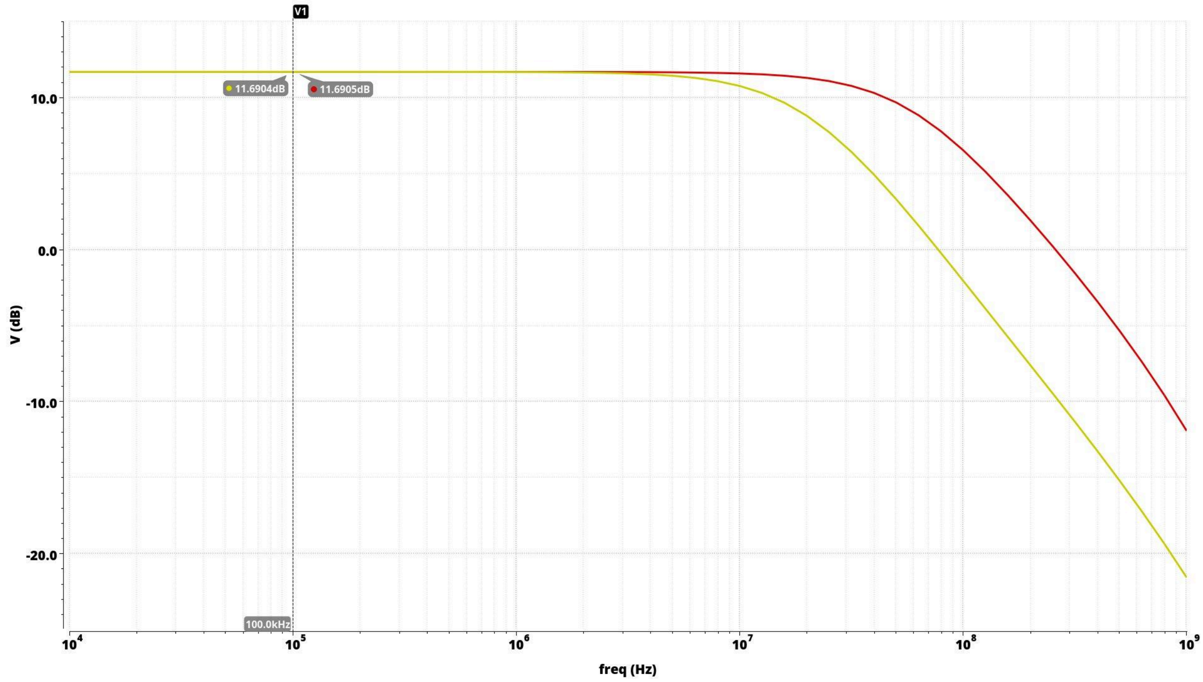


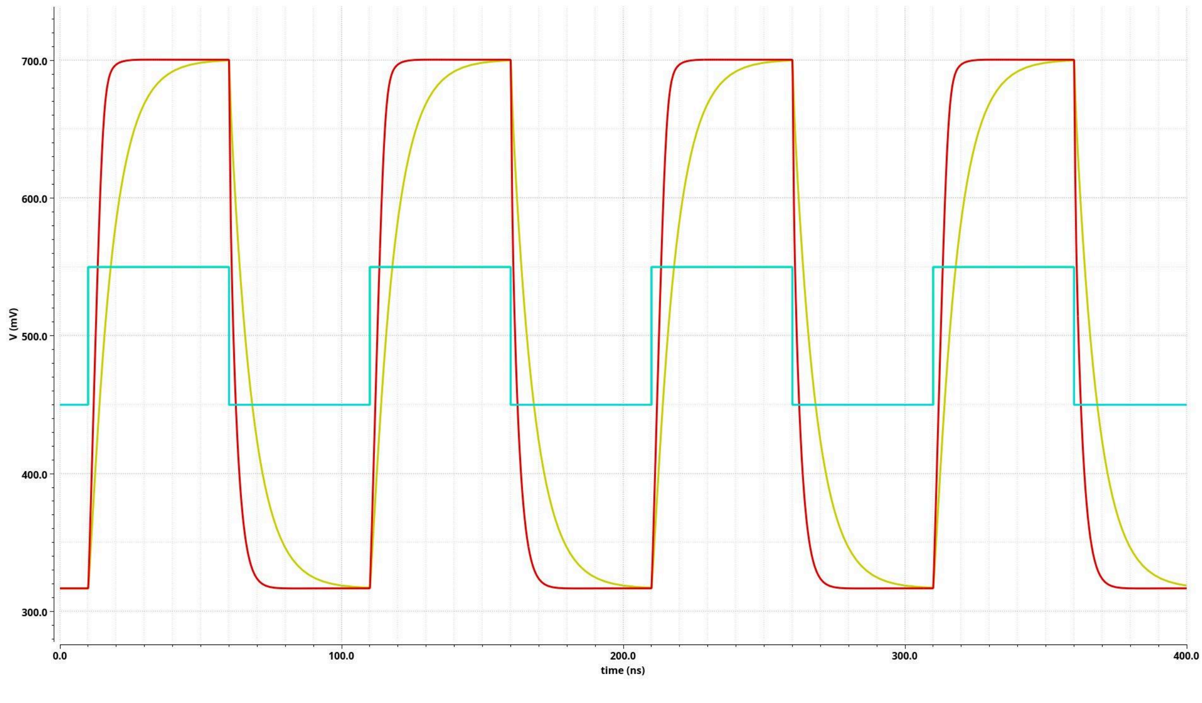
The result showed PM ≈ 92.4º > 75º which is quite close to the hand calculated value of 91.9º. Since there is a lot more PM headroom, the initial guess of C_C is too big and can be reduced to push the closed loop BW (which is the same as unity loop gain BW) higher at the cost of PM being closer to the 75 deg requirement.



The final C_C and R_C chosen are 0.72 pF and 1260 □, pushing the closed loop BW from 19.7 MHz to 61.8 MHz while PM ≈ 91.4º > 75º. Though the PM headroom is still more than sufficient, increasing the dominant pole further lead to the second pole appearing before unity gain BW.

The effect of increasing closed loop BW is clear in the following plots of closed loop AC frequency response and transient simulation:
Yellow trace: C_C = 2.5 pF, R_C = 475.9 Ω
Red trace: C_C = 0.72 pF, R_C = 1260 Ω





At low frequency,
open loop gain = closed loop gain + loop gain
open loop gain ≈ 11.7 dB + 27.4 dB
open loop gain ≈ 39.1 dB
which is close to the 41.7 dB predicted in part 1